Model quantities to an exact all-state certificate

1 Fixed candidate quantities

Freeze $z(0),\ h,\ \phi,\ a$ along the path

$c \in \mathbb{R}^5_+;\ d, \delta \in \mathbb{R}^4_+$

$\delta_i = w_i - w_0;\ w_0, b$ are scalars

Tilt may differ across candidates

2 Scalar derivative

$q_i = c_i z_i / (c^\top z), \quad G = w_0 + \delta^\top q$

$-\dot{G} = F(q) = a_{\text{aux}}^\top q - (\delta^\top q)(e^\top q)$

$(a_{\text{aux}})_i = \delta_i d_i, \quad e_i = d_i(1 - c_0/c_i)$

Append zeros: $\delta_0 = (a_{\text{aux}})_0 = e_0 = 0$

3 Six active-pair tests

$A_i = c_0 \delta_i d_i / c_i$

$K_{ij} = (\delta_i - \delta_j)(d_i - d_j)$

$B_{ij} = K_{ij} + c_0(\delta_i d_j / c_j + \delta_j d_i / c_i)$

$\mu_{ij} = B_{ij} + 2\sqrt{A_i A_j} \ \geq \ 0$

$K = 5$: four active states, six pairs

4 Certificate scope

All $\mu_{ij} \geq 0$: safe for every $z(0)$

One $\mu_{ij} < 0$: unsafe edge witness

Fix tilt, rates, readout and affinity

Changing context: recheck the score

All-state theorem: numerical trajectories remain diagnostics.